Q08: Scaling Limits and Forecasts

What a fitted curve can actually tell us

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The question

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Will performance saturate or models forget as parameters grow? Can we predict more-parameter effects before training?

Scaling laws are conditional models

$$L(N) \approx L_\infty + aN^{-\alpha}$$

- Fit observed loss/resource relations.
- Useful within the measured recipe and regime.
- Not a universal law of capability.

“More parameters” is incomplete

- Parameters N, data D, and compute C interact.
- A bigger model trained on too few tokens may be inefficient.
- Architecture, data quality, and objective also matter.

Kaplan et al. (2020); Hoffmann et al. (2022)

Forecast protocol

1. Run matched small-scale experiments.
2. Hold out a scale for validation.
3. Fit uncertainty, not only a line.
4. Declare metric and extrapolation range.
5. Check a separate downstream metric.

Saturation has several meanings

- Diminishing validation-loss improvement
- Benchmark ceiling or contamination
- Data-quality limit
- Inference/deployment budget limit

Diagnose the limiting mechanism before choosing an intervention.

Forgetting also has several meanings

- Catastrophic forgetting after task shift
- Example-level forgetting during training
- Memorization of rare or duplicated data

None follows automatically from parameter count.

Toy experiment

- Fit a power law to a stable synthetic regime.
- Check an interpolation point.
- Introduce a regime shift and inspect extrapolation error.

A fit is evidence about its data-generating regime.

Takeaway

Use scaling laws as budgets, not oracles
State the allocation, measured metric, range, uncertainty, and regime assumptions. Evaluate downstream behavior separately.